Design and Control of a Methyl Acetate Process Using Carbonylation of Dimethyl Ether

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The production of methyl acetate from the raw materials of methanol and carbon monoxide offers a potentially attractive alternative to other routes. The process has two distinct reaction and separation steps. First, methanol is dehydrated to form dimethyl ether (DME). Then carbon monoxide is combined with DME to produce methyl acetate. Each section involves reactors and distillation columns with either liquid or gas recycle. The purpose of this paper is to develop the economically optimum design of the two-step process and to study its dynamic control. The economics consider capital costs (reactors, distillation columns, heat exchangers, and compressors), energy costs (compressor work, reboiler heat input, and condenser refrigeration), the value of steam produced from the exothermic reactions, the cost of the carbon monoxide, and the heating value of a vent stream that is necessary for purging off the inert hydrogen that enters in the fresh carbon monoxide feed. The process illustrates a number of important design trade-offs among the many design optimization variables: reactor temperatures, reactor pressures, distillation column pressures, reactor sizes and purge composition. A plantwide control structure is developed for the entire two-section process that is capable of effectively handling large disturbances in production rate and fresh feed compositions.

1. Introduction

Methyl acetate is a commodity chemical with many uses. There are several processes used for its production. Methyl acetate is conventionally made by esterification of acetic acid with methanol over an acid catalyst. But, there is an alternative route in which methanol is first dehydrated to dimethyl ether and then carbonylated directly to methyl acetate. For a plant with carbon monoxide available on site, this may be a viable option. Recent advances in carbonylation catalysis promise improved economics for this approach.

This paper explores this alternative route that consists of two reaction steps: dehydration of methanol to form DME, followed by carbonylation with CO to form methyl acetate. As a point of reference, the price found on the ICIS.com Web site for methanol is \$1 to \$2 per gal (\$21.4 per kmol) and for methyl acetate is \$0.83 per lb (\$135 per kmol), which indicate a profitable process is possible.

In the following sections we develop reasonable conceptual designs of these two sections of the plant. Then plantwide dynamic control of the entire process is considered.

2. Dehydration Section

The dehydration reaction of methanol to form dimethyl ether and water can be conducted in either the vapor or the liquid phases. The kinetics used in this paper are discussed in detail in a later section. A vapor-phase tubular reactor is used. The reaction is exothermic.

There are many alternative reactor configurations. We consider three possible flowsheets:

1. One adiabatic reactor
2. Two adiabatic reactors with intermediate cooling
3. Cooled reactor

The most economically attractive of these three is shown in a later section of the paper to be the cooled reactor.

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2.1. Process Description of Dehydration Section. Figure 1 shows equipment and stream details of the optimum cooled reactor process. Fresh methanol is fed to a vaporizer at 500 kmol/h. The composition of the fresh feed is 99 mol % methanol and 1 mol % water. A recycle stream of methanol at 102.7 kmol/h is also fed to the vaporizer, which operates at 13.5 atm. This pressure is set by the downstream distillation column pressure of 10 atm, which is determined by the vapor pressure of DME at 318 K so that cooling water can be used in the condenser. The vaporizer temperature is 423 K, which requires medium-pressure steam (11 atm, 457 K). Vaporizer energy consumption is 5.149 MW.

The vapor stream is preheated in a feed-effluent heat exchanger (FEHE) to 628 K before entering the cooled tubular reactor. The reactor has 300 tubes (0.0245 m diameter and 10 m length). The Aspen heat-transfer model with "*constant medium temperature*" is used for this cooled tubular reactor. The optimum medium temperature is 665 K.

The exothermic heat of reaction is used to generate high-pressure steam on the shell side of the reactor (42 atm), which provides a credit for the economics of the process at \$9.88 per GJ. An overall heat-transfer coefficient of 0.28 kW K^{-1} m^{-2} is used. The per-pass conversion of methanol is 82.9%. The catalyst has a density of 2500 kg/m^3 with a void fraction of 0.4. The catalyst price is \$50 per kg.

The hot reactor effluent leaves at 658 K and is fed to the FEHE to preheat the reactor feed. The FEHE transfers 2.198 MW and cools the vapor stream to 467 K. An overall heat-transfer coefficient of 0.17 kW K^{-1} m^{-2} is assumed. The stream is cooled further in a heat exchanger (economizer) that preheats the fresh and recycled methanol streams to 400 K (overall heat-transfer coefficient = 0.28 kW K^{-1} m^{-2}). The stream then enters a water-cooled heat exchanger that cools it to 351 K before entering the first distillation column (overall heat-transfer coefficient = 0.85 kW K^{-1} m^{-2}.

Column C1 has 22 stages and is fed on Stage 12. We use the Aspen terminology in which Stage 1 is the condenser. The separation is an easy one, so few trays are required and the

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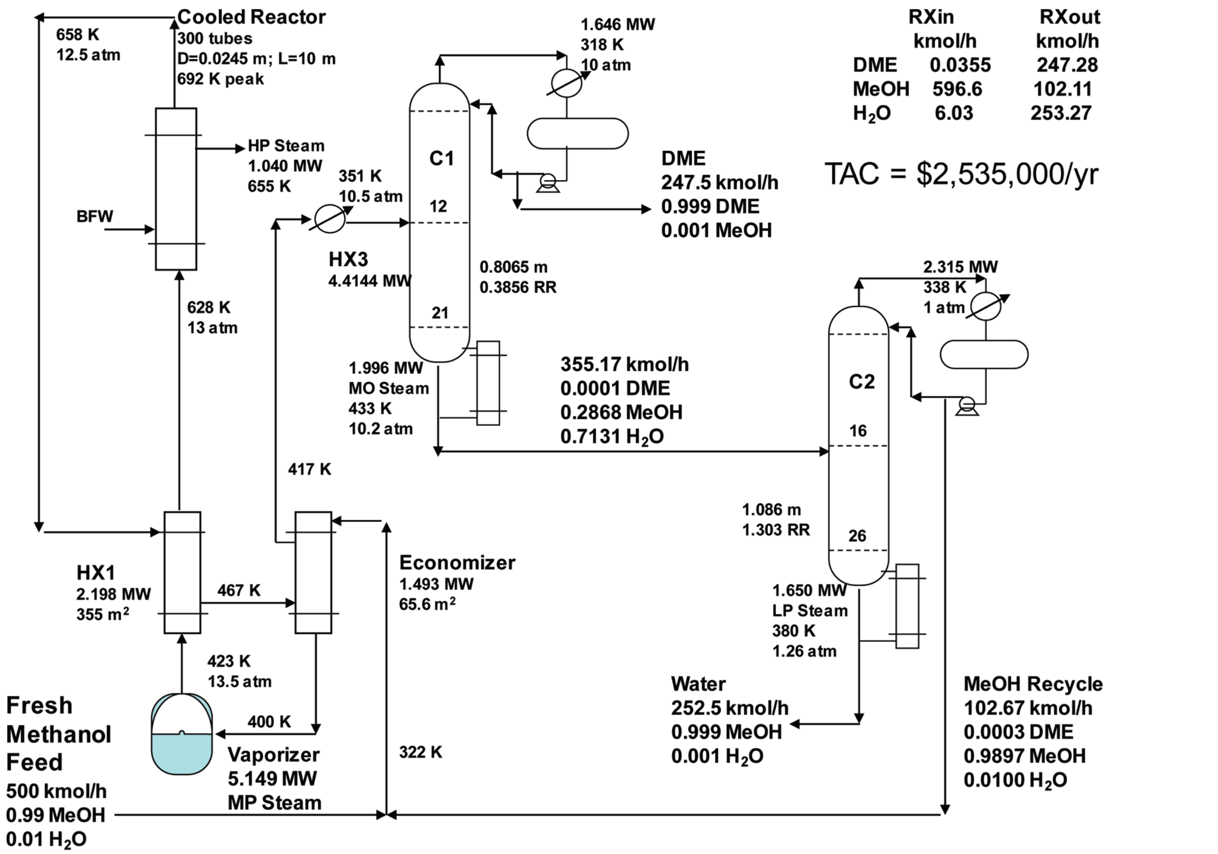

Figure 1. Dehydration flowsheet; cooled reactor (300 tubes).

reflux ratio is only 0.38. Reboiler duty is 1.996 MW using medium-pressure steam since the base temperature is 433 K. The distillate product is 99.9 mol % DME with 0.1 mol % methanol. This liquid distillate stream is pumped up to the pressure required in the carbonylation section of the plant. The bottoms stream is a mixture of methanol and water, which is fed to Column C2 for separation.

Column C2 produces a high-purity methanol distillate (98.97 mol % methanol, 0.03 mol % DME, and 1 mol % water), which is recycled back to the vaporizer. The bottoms is high-purity water (99.9 mol %). The column operates at 1 atm, using cooling water in the condenser and low-pressure steam in the reboiler (1.65 MW).

2.2. Dehydration Kinetics. The chemistry of the dehydration of methanol produces DME and water in a vapor-phase reaction.

$$2CH_3OH \leftrightarrow CH_3OCH_3 + H_2O \tag{1}$$

The reaction is exothermic, so the chemical equilibrium constant decreases with increasing temperature. Low-temperature operation can produce high conversion, but a large reactor will be required because the specific reaction rates are small. Thus there is a classical trade-off between reactor size and recycle in this kinetic system.

The forward reaction rate was taken from a rate expression found in a design project on the West Virginia University Chemical Engineering Department's Web site.[1] This project writeup cites Bandiera and Naccache[2] who measured the methanol dehydration rate over dealuminated H-mordenite catalyst at temperatures ranging 473−573 K. Under these conditions, Bandiera and Naccache found the rate to be first order in methanol partial pressure. They also found evidence of product inhibition which they interpreted as DME adsorption, and the rate constant in the WVU writeup corresponds to inhibition at an average DME mol fraction of 0.055 given their reactor pressure of 16.8 atm and an assumed catalyst bulk density of approximately 1000 kg/m^3. At our reactor pressures of 10−13 atm, this corresponds to an average DME mol fraction of 0.07−0.095, which is realistic if not exact.

In this work, the WVU rate expression (having units per unit bed volume) was converted back to a per unit bed mass basis assuming a bed bulk density of 1000 kg/m^3. The equilibrium constant for the reaction was estimated from known thermodynamic properties so that the final rate expression had the following form:

$$\begin{aligned} r_{dehy} &= k_{dehy}\left(p_{MeOH} - \frac{p_{DME}p_W}{p_{MeOH}K_{dehy}}\right) \\ r_{dehy} &= \text{MeOH dehydration rate, kmol/kg}_{\text{catalyst}}\text{/s} \\ k_{dehy} &= \text{dehydration rate constant, kmol/kg}_{\text{catalyst}}\text{/s/Pa} \\ K_{dehy} &= \text{dehydration equilibrium constant} \end{aligned} \tag{2}$$

With:

$$\begin{aligned} \ln k_{dehy}(\text{kmol/kg}_{\text{catalyst}}\text{/s/Pa}) &= -8.00 - 9680/T(\text{K}) \\ \ln K_{dehy} &= -2.8086 + 3061/T(\text{K}) \end{aligned} \tag{3}$$

Table 1 gives the kinetic and adsorption parameters entered into the Aspen LHHW reaction model to implement these kinetics.

2.3. Alternative Flowsheets. The flowsheet shown in Figure 1 is for a cooled reactor. Adiabatic reactors were also studied. The separation section is essentially the same in all flowsheets with small differences in reboiler duties for different methanol recycle flow rates.

2.3.1. One Adiabatic Reactor. Figure 2 shows the effects of changing reactor size (DR) and reactor inlet temperature (Tin) on the amount of methanol recycle required when one adiabatic

Table 1. Kinetic LHHW Parameters

Dehydration ($CH_3OH \rightarrow$ ½ CH3-O-CH3 + ½ H_2O)

Kinetic Factor: k = 0.000336; E = 80,480 kJ/kmol

Driving-Force Expressions

- Term 1
  - Conc. exponents for reactants: CH_3OH = 1
  - Conc. exponents for products: DME = 0; H_2O = 0
  - Coefficients: A = 0; B = C = D = 0
- Term 2
  - Conc. exponents for reactants: CH_3OH = −1
  - Conc. exponents for products: DME = 1; H_2O = 1
  - Coefficients: A = 2.8086; B = −3061; C = D = 0

Adsorption Expression

- Adsorption term exponent = 1
- Concentration exponents: Term 1: H_2O = 0; Term 2: H_2O = 0
- Adsorption constants: Term 1: A = -0.61395, B = 0, C = 0, D = 0; Term 2: A = -0.61395, B = 0, C = 0, D = 0

Carbonylation (CO + CH3-O-CH3 $\rightarrow$ CH3-CO-O-CH3)

Kinetic Factor: k = 1.7×10^{-12}; E = 1663 cal/mol; To = 473.15 K

Driving-Force Expressions

- Term 1
  - Conc. exponents for reactants: CO = 1; DME = 1
  - Conc. exponents for products: MeOAc = 0
  - Coefficients: A = B = C = D = 0
- Term 2
  - Conc. exponents for reactants: CO = 0; DME = 0
  - Conc. exponents for products: MeOAc = 0
  - Coefficients: A = −20; B = C = D = 0

Adsorption Expression

- Adsorption term exponent = 1
- Concentration exponents: Term 1: DME = 0; Term 2: DME = 1
- Adsorption constants: Term 1: A = B = C = D = 0; Term 2: A = B = C = D = 0

reactor is used. A reactor aspect ratio of LR/DR = 10 is assumed, so a DR = 0.5 corresponds to a reactor with diameter 0.5 m and length 5 m. For each reactor size, there is an optimum reactor inlet temperature that minimized methanol recycle. This is equivalent to maximizing per-pass conversion of methanol. Larger reactors require less recycle because the lower temperatures produce larger chemical equilibrium constants.

Figure 3 shows the flowsheet using one adiabatic reactor. The economic optimum size vessel has a diameter of 0.7 m and a length of 7 m. The fresh and recycle methanol are preheated in an economizer and vaporized. The vapor stream is preheated to 570 K in a FEHE, and leaves the adiabatic reactor at 686 K. The per-pass conversion is 81.7%. The flow rate of recycle methanol is 111.4 kmol/h.

The hot reactor effluent goes to a steam generator that produces high-pressure steam (1.194 MW), which is an economic credit. The stream is then cooled in three heat exchangers in series before entering the first distillation column.

2.3.2. Two Adiabatic Reactors. Figure 4 shows the flowsheet using two adiabatic reactors in series with intermediate cooling. The advantage of having two reactors with interstage cooling is that higher reactor inlet temperatures can be used to obtain faster kinetics, while at the same time having lower reactor exit temperatures, which give a higher chemical equilibrium constant. We assume that each reactor is the same size. The economic optimum size of each vessel is a diameter of 0.5 m and a length of 5 m.

After the vaporizer and FEHE, a small fired heater is required to attain the 620 K inlet temperature to the first reactor. Notice that this is higher than the reactor inlet temperature in the one-adiabatic reactor flowsheet (570 K). The effluent from this reactor leaves at 728 K and is cooled to 620 K by generating high-pressure steam in an intermediate heat exchanger. The cooled stream is fed to the second reactor in which a small additional conversion of methanol occurs, giving an exit temperature of only 625 K. The per-pass conversion through the two reactors is 86.2%. The flow rate of recycle methanol is 97.23 kmol/h, which should be compared to the methanol recycle flow rate of 111.4 kmol/h in the one-adiabatic flowsheet.

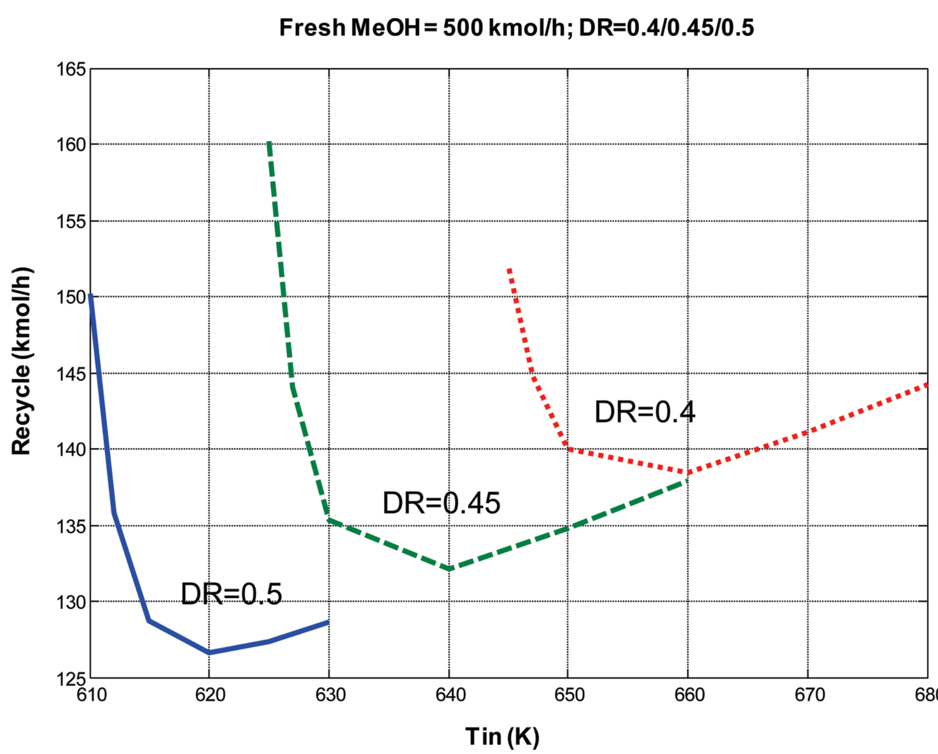

Figure 2. Effect of inlet temperature and reactor size; dehydration; one adiabatic.

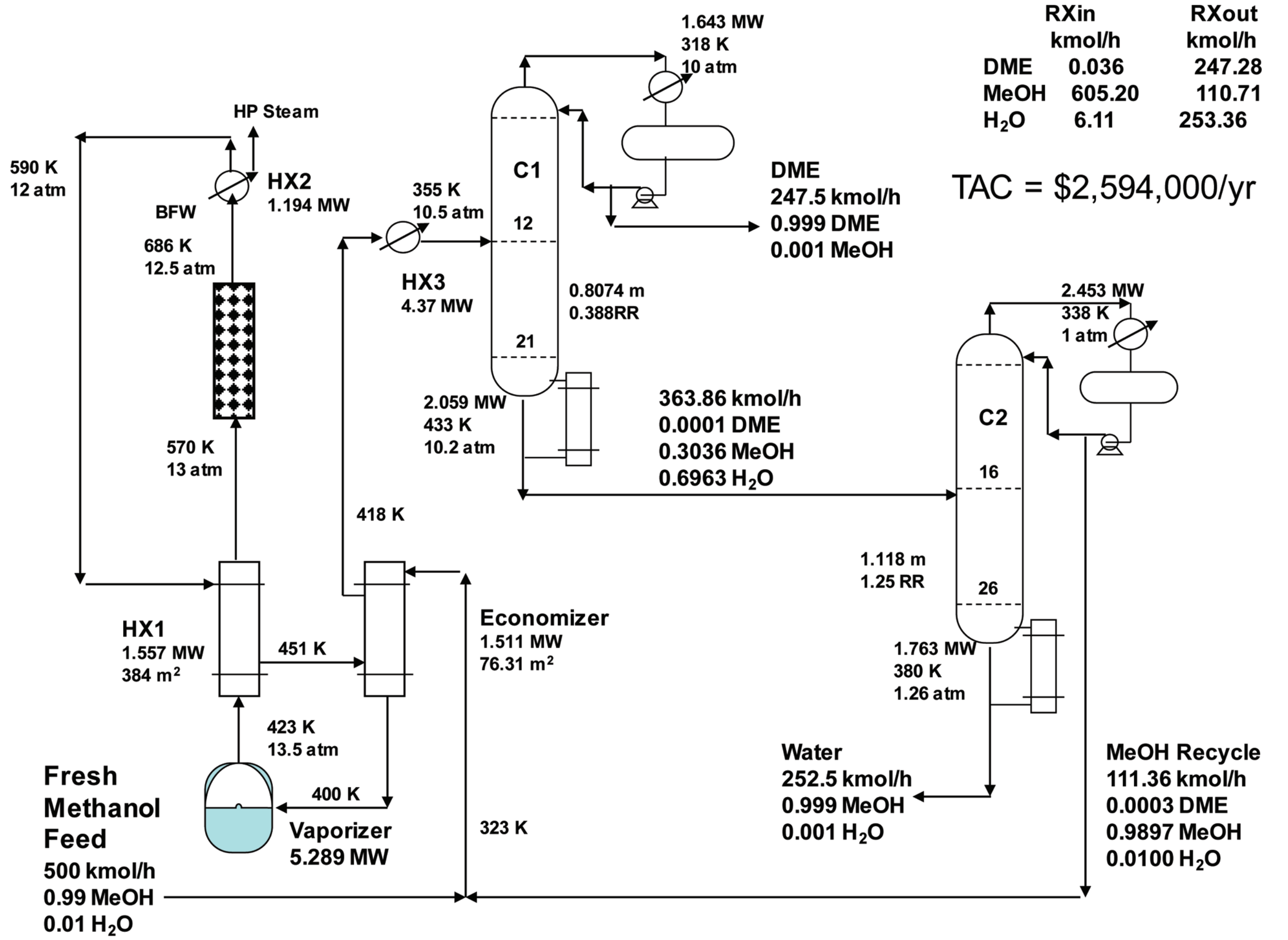

Figure 3. Dehydration: one adiabatic reactor (0.7 × 7 m).

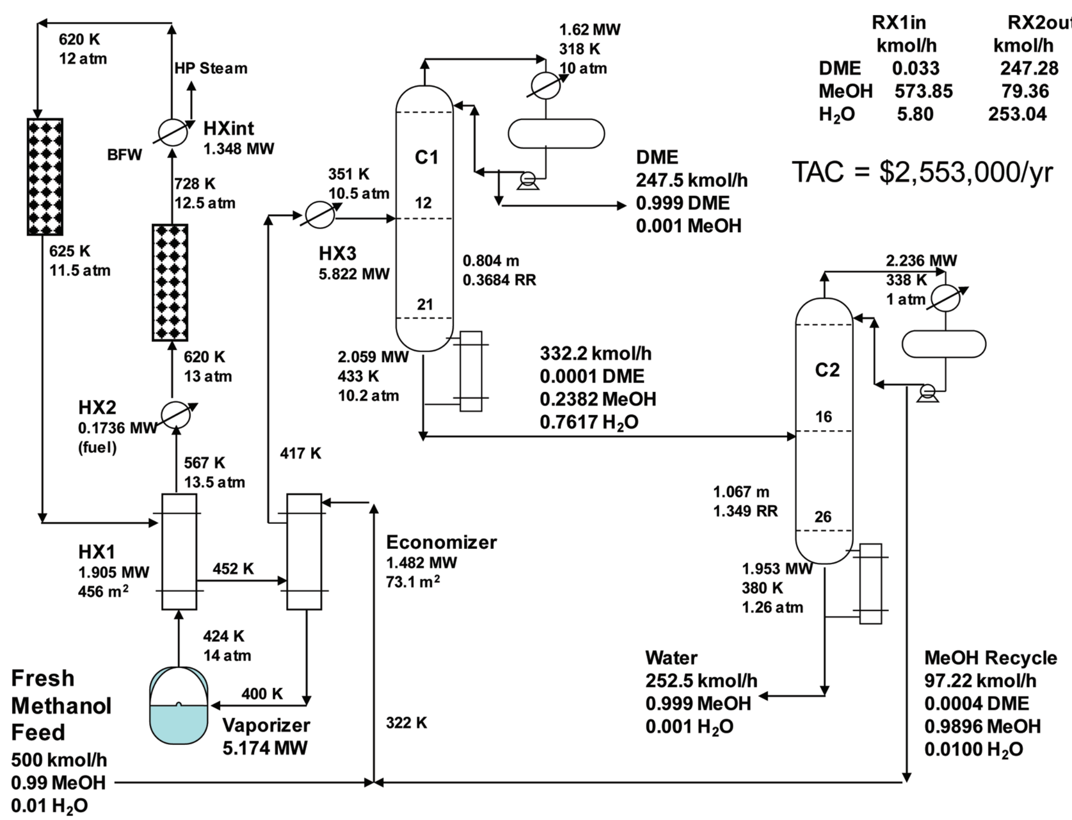

Figure 4. Dehydration: two adiabatic reactors (0.5 × 5 m).

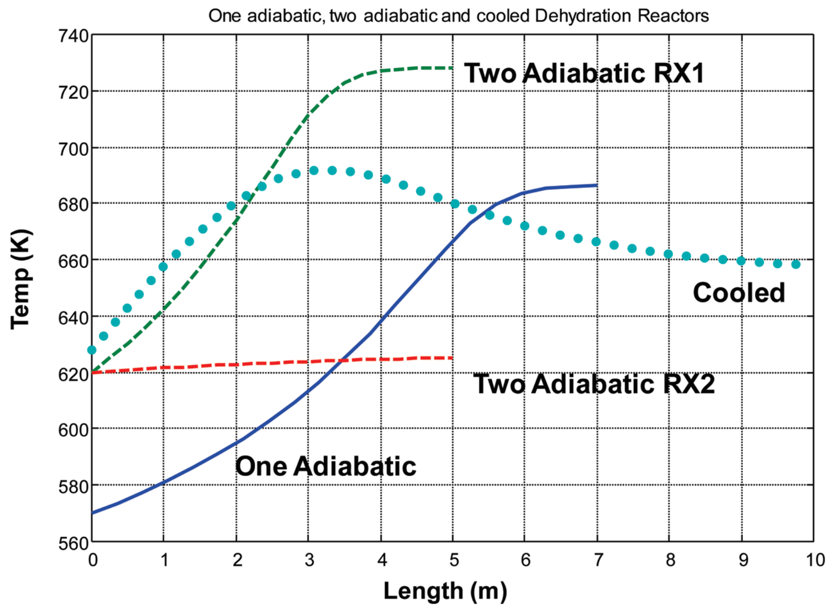

Figure 5. Dehydration; temperature profiles.

2.3.3. Cooled Reactor. This flowsheet has been described in detail in Section 2.1 and is shown in Figure 1. Note that the reactor cooling medium temperature is 655 K, which gives a reactor exit temperature of 658 K (and a peak temperature of 692 K). This exit temperature is lower than those of the one-adiabatic reactor flowsheet (668 K) and the first reactor in the two-adiabatic reactor flowsheet (728 K), which gives a larger chemical equilibrium constant. The methanol recycle flow rate is 102.7 kmol/h, which is intermediate between the one-adiabatic and two-adiabatic flowsheet recycle flow rates.

Figure 5 compares the temperature profiles through the reactors in the three alternative flowsheets. The one adiabatic reactor must have a low inlet temperature (570 K) and a large methanol recycle flow rate (111.36 kmol/h) to keep the reactor exit temperature from going too high (686 K).

In the two-adiabatic flowsheet, the inlet temperatures can be higher (620 K) than in the one-adiabatic flowsheet. The first reactor has a high exit temperature because of the high inlet temperature and because the methanol recycle flow rate is lower (97.23 kmol/h). But the low temperatures in the second reactor raise the chemical equilibrium constant and permit higher methanol conversion.

The cooled reactor with a 655 K cooling medium temperature has a peak temperature of 692 K at about 3 m down the reactor, as shown in the dotted curve in Figure 5. Conversion is not as high in the cooled reactor as in the two-adiabatic flowsheet because of the higher temperatures, so the recycle flow rate is higher (102.7 kmol/h). The economic analysis given in the next section shows that the cooled reactor flowsheet has the minimum total annual cost, primarily because less catalyst is needed.

2.4. Optimization of Three Flowsheets. The three flowsheets have similar but slightly different design optimization variables. Reactor size and reactor temperature are the dominant variables. The basis for cost and sizing calculations are given in Table 2. These parameters are taken from Douglas[3] and Turton et al.[4]

Total annual cost (TAC) is the economic criterion used for choosing the optimum design. The cost of energy in the vaporizer and column reboilers is reduced by the credit for steam produced (in the heat exchanger after the reactor in the one-adiabatic flowsheet, in the intermediate heat exchanger in the two-adiabatic flowsheet or in the reactor in the cooled reactor flowsheet). The annual cost of capital is the total capital investment divided by a 3-year payback period. Capital investment includes the vaporizer, economizer, FEHE, reactor(s), partial condenser, column vessels, column condensers, and column reboilers).

Table 2. Basis of Economics and Equipment Sizing

Column Diameter: Aspen Tray Sizing

Column Length: NT trays with 2 ft spacing plus 20% extra length

Columns, Separator and Adiabatic Reactor Vessels (Diameter and Length in meters)
Capital cost = 17,640 $(D)^{1.066}$ $(L)^{0.802}$

Condensers (Area in m^2)
Heat-transfer coefficient = 0.852 kW/K-m^2
Differential temperature: C1 and C2 = Reflux-drum temperature -310 K
C3 = 300 – 253 K
Capital cost = 7296 $(Area)^{0.65}$

Reboiler (Area in m^2):
Heat-transfer coefficient = 0.568 kW/K-m^2
Differential temperature: C1 (MP steam) = 457 – base temperature
C2 (LP steam) = 433 – base temperature
C3 (LP steam) = 433 – base temperature
Capital cost = 7296$(Area)^{0.65}$

Economizer, Feed-Effluent, Partial Condenser and Vaporizer Heat Exchanger (Area in m^2):
Capital cost = 7296$(Area)^{0.65}$
Heat-transfer coefficients: Economizer = 0.28 kW K^{-1} m^{-2}
FEHE = 0.17 kW K^{-1} m^{-2}
Partial condenser = 0.852 kW K^{-1} m^{-2}
Vaporizer = 0.85 kW K^{-1} m^{-2}

Cooled Reactor Vessel (based on heat-transfer area of tubes)
Heat-transfer coefficient = 0.28 kW K^{-1} m^{-2}
Dehydration reactor; Cooling medium temperature = 655 K
Carbonylation reactor; Cooling medium temperature = 460 K
Capital cost = 7296 $(Area)^{0.65}$

Catalyst
Cost - \$50/kg , density = 2500 kg/m^3 , void fraction = 0.4

Energy cost
LP steam = \$7.78 per GJ
MP steam = \$8.22 per GJ
HP steam = \$9.88 per GJ
Electricity = \$16.6 per GJ

$$TAC = \frac{Capital\ Cost}{Payback\ Period} + Energy\ Cost$$

Payback period = 3 years

2.4.1. Single Adiabatic Reactor. For a single adiabatic reactor, the size of the reactor (represented by DR) and the reactor inlet temperature are the dominant variables. The flowsheet is optimized by selecting a reactor size and finding

Table 3. Optimum One-Adiabatic Reactor Flowsheet

	DR (m)		
	0.6	0.7	0.8
LR (m)	6	7	8
Tin (K)	590	570	555
Recycle D2 (kmol/h)	117.9	111.4	106.2
QR2 (MW)	1.849	1.763	1.696
Qvaporizer (MW)	5.345	5.289	5.244
Steam Value (10^6 \$/yr)	0.3674	0.3720	0.3755
Reactor Vessel (10^6 \$)	0.0659	0.0879	0.1128
Catalyst (10^6 \$)	0.2121	0.3367	0.5027
C1 Capital (10^6 \$)	0.6172	0.6149	0.6131
C2 Capital (10^6 \$)	0.5218	0.5085	0.4979
Total Capital (10^6 \$)	1.756	1.886	2.063
Energy Cost (10^6 \$/yr)	2.010	1.965	1.930
TAC (10^6 \$/yr)	2.596	**2.594**	2.618

the reactor inlet temperature that gives the minimum required methanol recycle flow rate. This minimizes vaporizer energy and separation cost in Column C2 in which the methanol is taken overhead and recycled. The total annual cost for each reactor size is evaluated and the minimum case is selected.

The total annual cost (TAC) considers both energy and capital costs. The annual cost of the energy comes from the duties in the vaporizer and in the two column reboilers. There is an energy credit for the high-pressure steam generated in the heat exchanger after the reactor. Capital costs include the capital cost of the heat exchangers (vaporizer, FEHE, economizer, partial condenser, column condensers and column reboilers) and distillation column vessels. The total annual cost is energy costs plus annual capital cost (total capital divided by a 3-year payback period).

Table 3 gives economic results for a single adiabatic flowsheet for three different reactor sizes (DR = 0.6, 0.7, and 0.8 m). For each case, the reactor inlet temperature T_{in} that minimizes the flow rate of recycle is found. The optimum inlet temperature decreases as reactor size is increased. Recycle flow rates, vaporizer duty, and C2 reboiler duty decrease as reactor size increases.

Naturally, the capital cost of the reactor vessel and the catalyst increase directly with reactor size. The capital cost of Column C1 changes very little, but the capital cost of Column C2 decrease slightly with increasing reactor size.

Total capital cost increases while total energy cost decreases as the reactor is made larger. The net effect is a minimum TAC of \$2,594,000 per year for the one-adiabatic reactor flowsheet when a 0.7 by 7 m reactor is used.

2.4.2. Two Adiabatic Reactors. We assume that the reactors are of equal size, but the inlet temperatures can be different. A range of reactor inlet temperatures to the two reactors is explored for a specified reactor size. The values of the two inlet temperatures that minimize recycle flow rate are found. In all the cases considered, it turned out that the inlet temperatures are the same. Then a different reactor size is specified, and the calculations are repeated. The reactor size and set of reactor inlet temperatures that give the minimum total annual are selected.

Table 4 gives economic results for the two-adiabatic reactor flowsheet for three different reactor sizes (DR = 0.4, 0.5, and 0.6 m). These are the dimensions of each of the two reactors. The optimum inlet temperatures decrease as reactor size is increased, as do recycle flow rates, vaporizer duty, and C2 reboiler duty. The capital cost of the two reactor vessels is somewhat higher than the capital cost of the larger single reactor in the one-adiabatic flowsheet. However, the cost of the catalyst is smaller since the total reactor volume is smaller in the two vessels than in the single larger vessel. The capital cost of Column C1 is about same in the two flowsheets, but the capital cost of Column C2 is somewhat smaller in the two-adiabatic flowsheet because of the smaller recycle flow rate.

Table 4. Optimum Two-Adiabatic Reactor Flowsheet

	DR (m)		
	0.4	0.5	0.6
LR (m)	4	5	6
Tin (K)	650	620	575
Recycle D2 (kmol/h)	107.1	97.23	106.2
QR2 (MW)	1.724	1.593	1.696
Qvaporizer (MW)	5.260	5.174	5.244
Steam Value (10^6 \$/yr)	0.4153	0.4191	0.4191
Two Reactor Vessels (10^6 \$)	0.0618	0.0937	0.1318
Catalyst (10^6 \$)	0.1257	0.2454	0.4241
C1 Capital (10^6 \$)	0.6153	0.6117	0.6091
C2 Capital (10^6 \$)	0.5014	0.4803	0.4648
Total Capital (10^6 \$)	1.709	1.842	2.044
Energy Cost (10^6 \$/yr)	2.009	1.939	1.888
TAC (10^6 \$/yr)	2.579	**2.553**	2.569

Table 5. Optimum Cooled Reactor Flowsheet

	tubes		
	200	300	400
LR (m)	10	10	10
Tube ID (m)	0.0245	0.0245	0.0245
Tmedium (K)	665	655	645
Recycle D2 (kmol/h)	109.9	102.7	98.14
QR2 (MW)	1.745	1.650	1.592
Qvaporizer (MW)	5.276	5.214	5.174
Steam Value (10^6 \$/yr)	0.3190	0.3240	0.3276
Reactor Vessel (10^6 \$)	0.1973	0.2568	0.3095
Catalyst (10^6 \$)	0.1267	0.1900	0.2534
C1 Capital (10^6 \$)	0.6137	0.6109	0.6092
C2 Capital (10^6 \$)	0.5056	0.4905	0.4811
Total Capital (10^6 \$)	1.651	1.754	1.858
Energy Cost (10^6 \$/yr)	2.000	1.950	1.918
TAC (10^6 \$/yr)	2.550	**2.535**	2.538

Total capital cost increases while energy cost decreases as the reactor is made larger. The net effect is a minimum TAC of \$2,553,000 per year for the two-adiabatic reactor flowsheet when two 0.5 by 5 m reactors are used. Note that this is somewhat smaller than the TAC of the one-adiabatic flowsheet.

2.4.3. Cooled Reactor. The cooled reactor capital cost is based on its heat-transfer area. The optimum design has 300 tubes with diameter 0.0245 m and length 10 m. High-pressure steam is generated in the reactor and is an energy credit. The number of tubes is specified and the temperature of the cooling medium is varied to find the value that gives the minimum recycle flow rate.

Table 5 gives designs with different numbers of tubes. The optimum cooling-medium temperature and the methanol recycle flow rate decrease with increasing number of tubes. The cost of the reactor vessel and the catalyst increase as more tubes are used. Notice that the cost of the vessel (\$256,800 for the tube-in-shell reactor) for the cooled reactor with the optimum 300 tubes is much larger than the two simple vessels in the two-adiabatic flowsheet (\$93,700). However, the catalyst cost is lower (\$190,000 versus \$245,400) since the cooled reactor requires the smallest amount of catalyst.

Total capital cost increases and total energy cost decreases as more reactor tubes are used. The minimum TAC of \$2,535,000 occurs with 300 tubes. The cooled reactor flowsheet has the smallest TAC.

Figure 6 give direct comparisons among the three flowsheets, showing how methanol recycle, reactor temperatures, energy costs, and TAC vary with the total catalyst. The cooled reactor

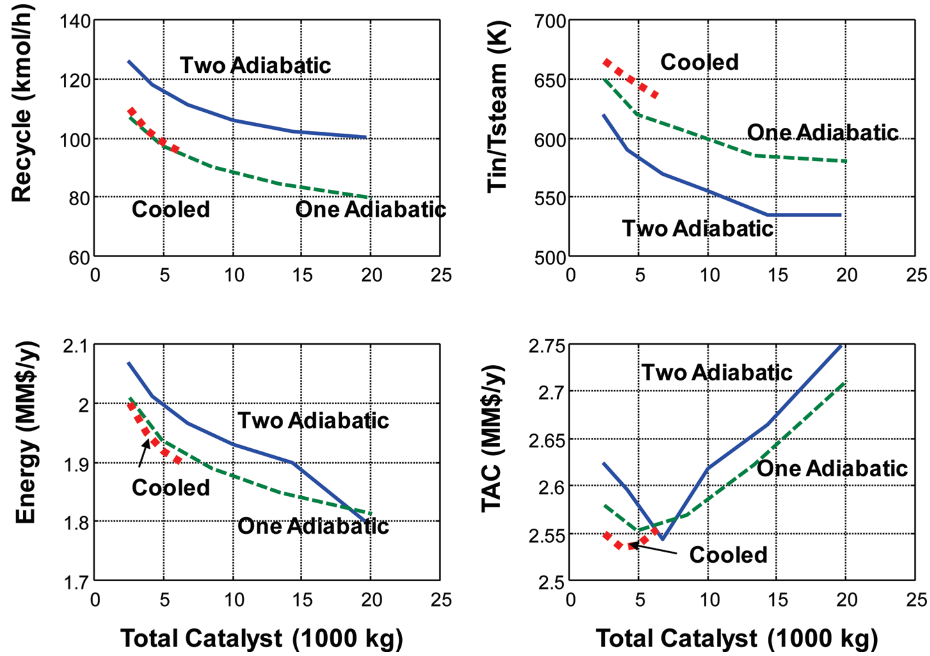

Figure 6. Alternative reactor design.

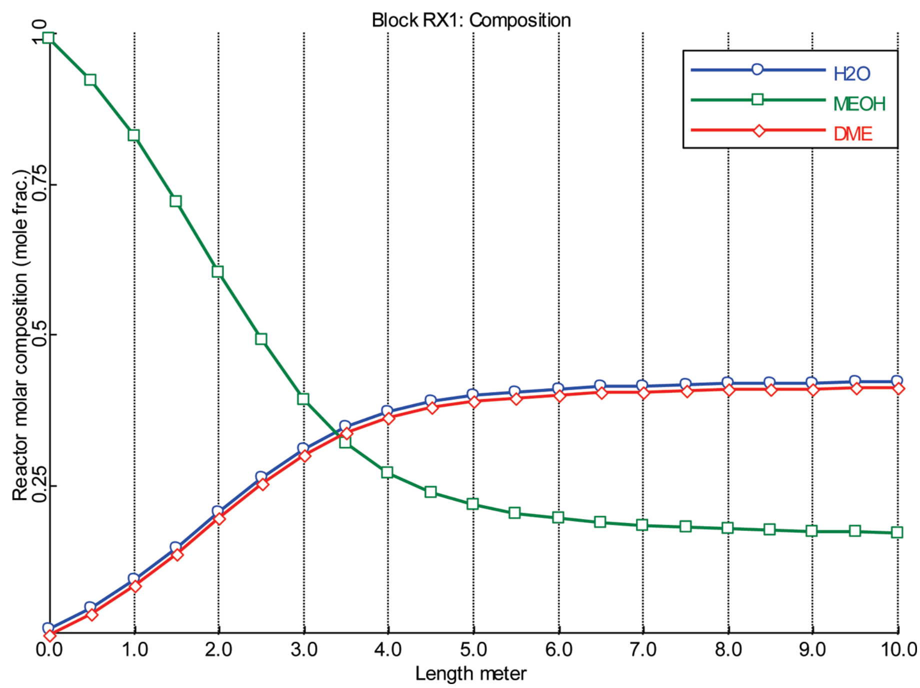

Figure 7. Cooled reactor (300 tubes) composition profiles.

is selected for the dehydration section of the plant. Figure 7 shows the composition profiles down the length of the cooled reactor.

2.4.4. Optimization of Distillation Columns. The two distillation columns in the dehydration section are optimized by finding the total number of trays in each column that minimize total annual cost. The feed flow rates and compositions found in the optimum cooled reactor design are used in these calculations. The specifications for distillate and bottoms purities are those previously discussed. The feed tray location is for each case that which minimizes reboiler duty.

Table 6 gives results of the economic analysis over a range of column sizes. Capital costs increase as more stages are added, while energy costs decrease. The optimum design for Column C1 is 22-stage column, fed on Stage 12. The optimum design for Column C2 is 27-stage column, fed on Stage 16.

In the following section, the steady-state design of the carbonylation section is developed. The last section of the paper considers the dynamic control of both sections coupled together.

3. Carbonylation Section

The liquid DME produced in the first section of the plant is fed into the carbonylation section. Carbon monoxide is available at 5 atm, so this gaseous fresh feed stream must be compressed to the pressure selected for operation of the carbonylation reactor. This makes system pressure a critically important design optimization variable. As discussed below, higher pressures favor reaction kinetics, so reactor size is reduced and less recycle

Table 6. Optimum Column Designs

	C1		
NT	17	22	27
NF	9	12	14
QR (MW)	2.064	2.019	2.018
ID (m)	0.8095	0.7978	0.7984
Total Capital (10^6 \$)	0.5782	0.6062	0.6418
Total Energy (10^6 \$/y)	0.5350	0.5233	0.5232
TAC (10^6 \$/y)	0.7278	**0.7254**	0.7371
	C2		
NT	22	27	32
NF	14	16	18
QR (MW)	1.695	1.650	1.649
ID (m)	1.098	1.086	1.084
Total Capital (10^6 \$)	0.4646	0.4905	0.5208
Total Energy (10^6 \$/y)	0.4159	0.4049	0.4046
TAC (10^6 \$/y)	0.5707	**0.5683**	0.5782

is required to attain high conversion. This saves capital investment in the reactor vessel, the catalyst, and the recycle compressors. However, higher system pressure means higher capital and energy costs for the fresh feed CO compressor that must increase the pressure from the supply pressure of 5 atm.

The fresh CO feed contains an impurity of 2 mol % hydrogen, which is assumed to be inert in the system and must be purged from the process. This makes the concentration of hydrogen in the gas recycle and purge stream another important design optimization variable. The higher the hydrogen concentration in the purge steam, the lower the losses of CO, DME, and methyl acetate in the purge stream. However, higher hydrogen concentrations require higher gas recycle flow rates and larger reactors to maintain conversion. The purge stream only has heating value.

The economic optimization of the process is discussed in a later section. The final design flowsheet features a reactor with 1000 tubes, an operating pressure of 30 atm, and a hydrogen impurity level of 40 mol %.

3.1. Process Description. Figure 8 shows the flowsheet of the carbonylation section of the process. The liquid DME from the first section of the plant is pumped to 32 atm and fed to a vaporizer, which operates at 372 K, so low-pressure steam can be used (6 atm, 433 K). The flow rate of the DME is fixed in the steady-state design of this section at 250 kmol/h and has a composition of 99.9 mol % DME and 0.1 mol % methanol. The vapor stream is combined with three gas streams: the fresh CO feed, a recycle gas stream from a separator tank, and a vapor distillate stream from the top of the reflux-drum of a distillation column.

The required flow rate of the fresh CO feed is 262 kmol/h, which is higher than the stoichiometric amount needed to react with the 250 kmol/h of DME because of the loss of CO in the purge stream. The gas is compressed from 5 to 32 atm in compressor K_{CO}, which requires 636 kW of compression work. Compressor work is priced at the cost of electricity (\$16.8 per GJ).

The total vapor stream is fed to a multitube tubular reactor that is packed with solid catalyst. The reactor has 1000 tubes, 0.05 m diameter, and 10 m length. The catalyst has a density of 2500 kg/m^3 and a void fraction of 0.4. The total weight of catalyst is 48,800 kg. At a cost of \$50 per kg, the catalyst cost is \$2,440,000. The capital cost of the reactor vessel is estimated from its heat-transfer area (1570 m^2) to be \$872,000.

The heat from the exothermic reaction is removed by generating low-pressure steam. A plug-flow reactor model in Aspen Plus is used with the cooling option of "*constant medium temperature*" at 460 K. The heat removal rate is 7.165 MW of energy, using an overall heat-transfer coefficient of 0.28 kW K^{-1} m^{-2}. The low-pressure steam generated in the reactor represents a credit of \$1,760,000 per year. The Ergun equation is used to calculate pressure drop through the reactor tubes. Total pressure drop through the gas loop is 2 atm.

The hot vapor stream from the reactor at 475 K is partially condensed in a water-cooled heat exchanger (4.369 MW) and fed to a separator tank, which operates at 320 K and 30 atm. Most of the gas from the separator is compressed back to 32 atm in compressor K_1, which requires 33.8 kW of compression work with a gas flow rate of 566.6 kmol/h.

A small portion of the separator gas is purged from the system to remove all the hydrogen entering in the fresh CO feed. The required purge flow rate is 13.1 kmol/h when the hydrogen composition is 40 mol %. The other components lost in the purge stream are CO (55.24 mol %) and small amounts of DME (2.16 mol %) and methyl acetate (2.59 mol %). As discussed later in this paper, lower hydrogen compositions result in higher losses of CO, which requires more fresh feed of CO. The purge stream represents a small credit to the operation since it has heating value. Using a value of \$6 per GJ and estimating the useful energy obtained from burning the gas in 5% excess air, the value of the purge stream is estimated to be \$1.6 per kmole with a purge gas composition of about 50/50 CO and hydrogen.

The liquid from the separator is fed on Stage 2 of a 17-stage distillation column operating at 5 atm. Aspen notation is used for tray numbering with the reflux drum being Stage 1. The separator liquid contains small amounts of dissolved light components (essentially no hydrogen but some CO and DME) that go overhead in the column and are recycled.

The column operating pressure of 5 atm is selected so that low-pressure steam can be used in the reboiler. The base temperature is 387 K when the bottoms methyl acetate product is 99.9 mol % methyl acetate and 0.1 mol % methanol. The condenser temperature is set at 300 K by including about 5 mol % methyl acetate in the vapor distillate. This means that refrigeration is required (at 253 K and \$7.89 per GJ). However, the condenser duty is fairly small (0.653 MW) since the vapor distillate is relatively small (52.39 kmol/h) and the reflux ratio is 1.32. The vapor distillate is compressed in compressor K_2 to 32 atm, requiring 101.6 kW of compression work.

The optimization of the column in terms of the number of trays and feed-tray location is discussed in a later section. The methyl acetate product is produced at a rate of 249.7 kmol/h and a purity of 99.9 mol %. The impurity is the small amount of methanol that comes in with the DME feed.

3.2. Carbonylation Kinetics. The chemistry of the carbonylation involves the reaction of carbon monoxide with dimethyl ether.

$$CO + CH_3OCH_3 \Leftrightarrow CH_3COOCH_3 \tag{4}$$

According to recent research (Cheung et al.[5]), DME can be carbonylated with high selectivity to form MeOAc. The reaction rate data given by Cheung, et al. may be reasonably represented under steady operating conditions by a rate expression of the following form derived from their mechanism:

$$\begin{gathered} r_{carbonyl} = k_{carbonyl}\left(\frac{p_{CO}}{1 + K_W p_W}\right) \\ r_{carbonyl} = \text{carbonylation, kmol/kg}_{\text{catalyst}}\text{/s} \\ k_{carbonyl} = \text{carbonylation rate constant, kmol/kg}_{\text{catalyst}}\text{/s/Pa}^{-1} \\ K_W = \text{H}_2\text{O adsorption equilibrium constant, Pa}^{-1} \end{gathered} \tag{5}$$

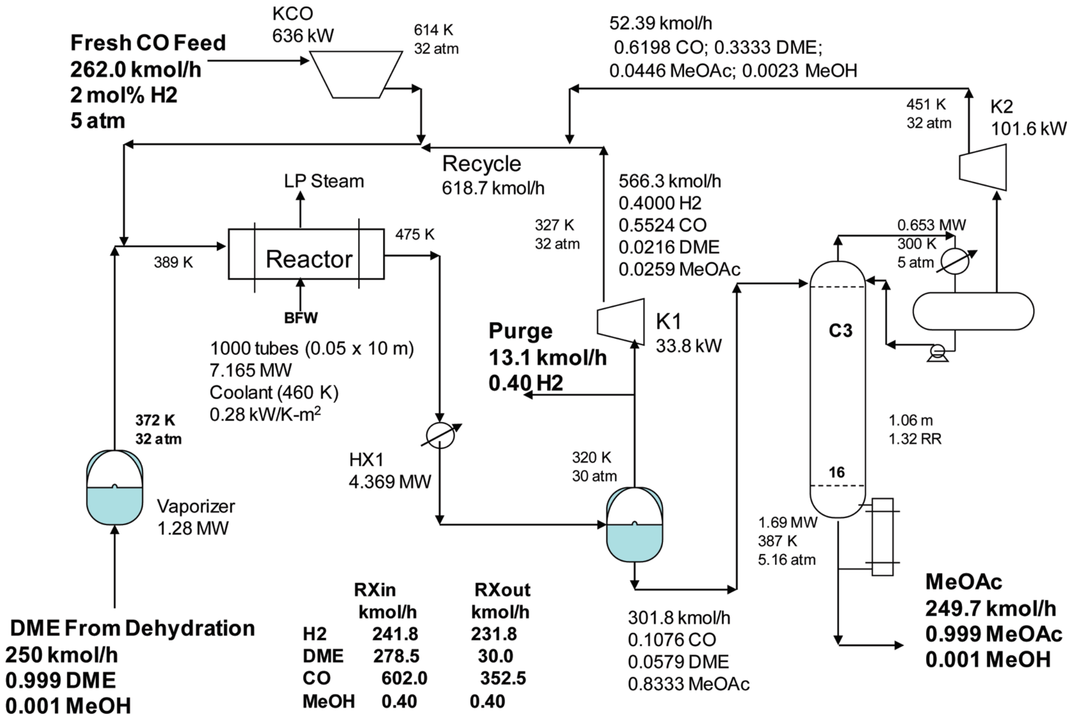

Figure 8. Carbonylation flowsheet; 30 atm; 1000 tubes; purge 40 mol % H2.

Data in Cheung et al. suggest that for DME-pretreated H-mordenite catalysts:

$$k_{carbonyl} = 8.2 \times 10^{-5} \exp\left(-\frac{8370}{T(\mathrm{K})}\right) \text{ kmol/kg}_{\text{catalyst}}\text{/s/Pa} \quad (6)$$

In practice, the inhibiting effect of water is so strong that it must be removed from the DME stream prior to feeding the carbonylation step. In this work it has been assumed that the DME feed stream is sufficiently dry that the water adsorption term can be neglected. That is reflected in the constants given in Table 1 for implementation in Aspen Plus.

An additional issue for simulation arises because this expression is zero order in one of the reactants, DME. In the event that DME is the limiting reactant and because the reaction is irreversible, this could result in calculated conversions of DME in excess of 100% (with corresponding negative DME outflows). To avoid this occurrence, the above expression was multiplied by a factor that is equal to unity except at very low DME partial pressures as shown below:

$$r_{carbonyl} = k_{carbonyl}\left(\frac{p_{CO}}{1 + K_W p_W}\right)\left(\frac{K_{DME} p_{DME}}{1 + K_{DME} p_{DME}}\right)$$
$$K_{DME} = \text{DME pseudo-adsorption equilibrium constant} = 1 \text{ Pa}^{-1} \quad (7)$$

This overall rate expression remains zero-order in DME up to very high conversions but approaches first-order behavior as 100% DME conversion is approached so that the rate goes to zero as DME approaches complete disappearance.

3.3. Effect of Parameters. Preliminary exploration of the effects of temperature and pressure on the carbonylation reaction reveal some important findings. A plug-flow reactor is modeled in Aspen Plus with 1500 tubes. The reactor is fed with 250 kmol/h of CO and 250 kmol/h of DME.

In the upper graph in Figure 9, the pressure is fixed at 30 atm, and the cooling-medium temperature is varied over a wide range. The methyl acetate produced is plotted versus temperature. Remember this is the temperature of the heat-removal medium low-pressure steam in the shell side of the reactor, not the reactor temperature, which varies down the length of the reactor. It is clear that higher temperatures favor methyl acetate production. The maximum cooling-medium temperature for effective operation of the catalyst is assumed to be 460 K. Therefore the cooling-medium temperature is set to this value.

In the lower graph in Figure 9, the cooling-medium temperature is fixed at 460 K, and the pressure is varied over a wide range. The methyl acetate produced is plotted versus pressure. It is clear that higher pressures favor methyl acetate production. However, the higher the pressure, the more compression work is required in the fresh CO compression. So the optimum system pressure involves an economic trade-off between compression costs and reactor size.

There is an additional consideration that can limit reactor pressure. The reaction is a gas phase reaction. Therefore the pressure in the reactor cannot be so high that liquid is formed in the reactor at the selected cooling-medium temperature of 460 K. The vapor fraction of the reactor effluent stays at unity up to a pressure of 40 atm. At 42 atm, the vapor fraction is 0.831. At 45 atm, it is 0.588. Therefore, a conservative maximum pressure of 30 atm is used in the design. In the next section, an economic evaluation will demonstrate that this maximum 30 atm operating pressure is the optimum. The cost of additional compression is not as large as the savings in purge losses and lower recycle flow rates.

The second preliminary exploratory study looked at the effects of excess amounts of the two reactants to see if the reactor should operate with excess CO or with excess DME to drive the reaction to higher conversion to methyl acetate. Figure 10 shows how the production of methyl acetate changes as the flow rate of either CO or DME fed to the reactor is varied with the other reactant feed fixed. A reactor with 1500 tubes, operating at 30 atm with a cooling-medium temperature of 460 K, is used in all these cases. The dashed line gives results for the case when the flow rate of DME is fixed at 250 kmol/h, and the flow rate of CO is varied over a wide range. It is clear that

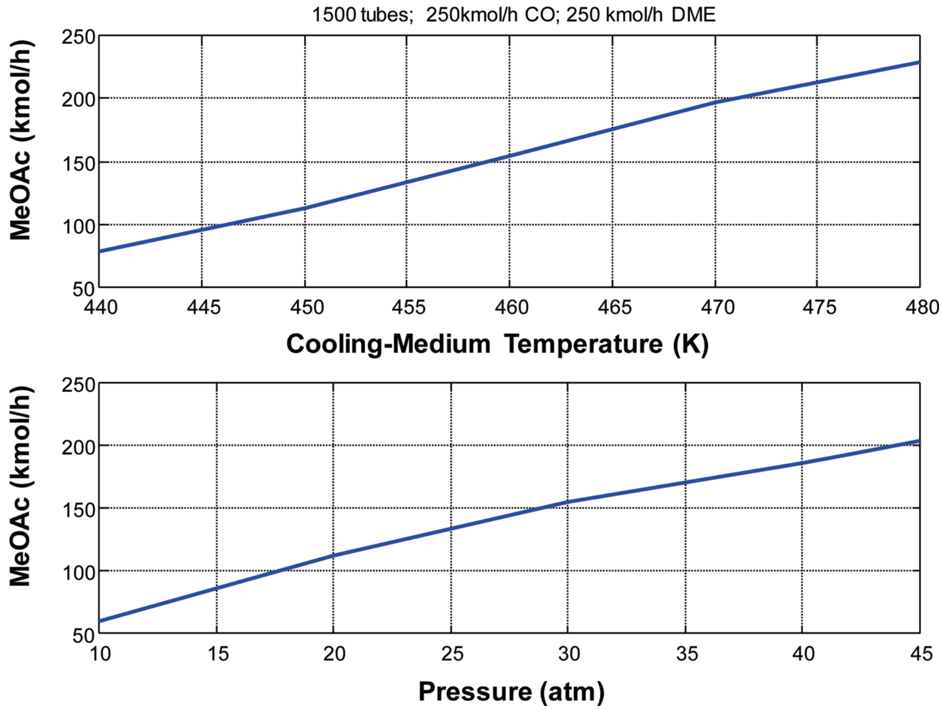

Figure 9. Carbonylation kinetics; effect of temperature and pressure.

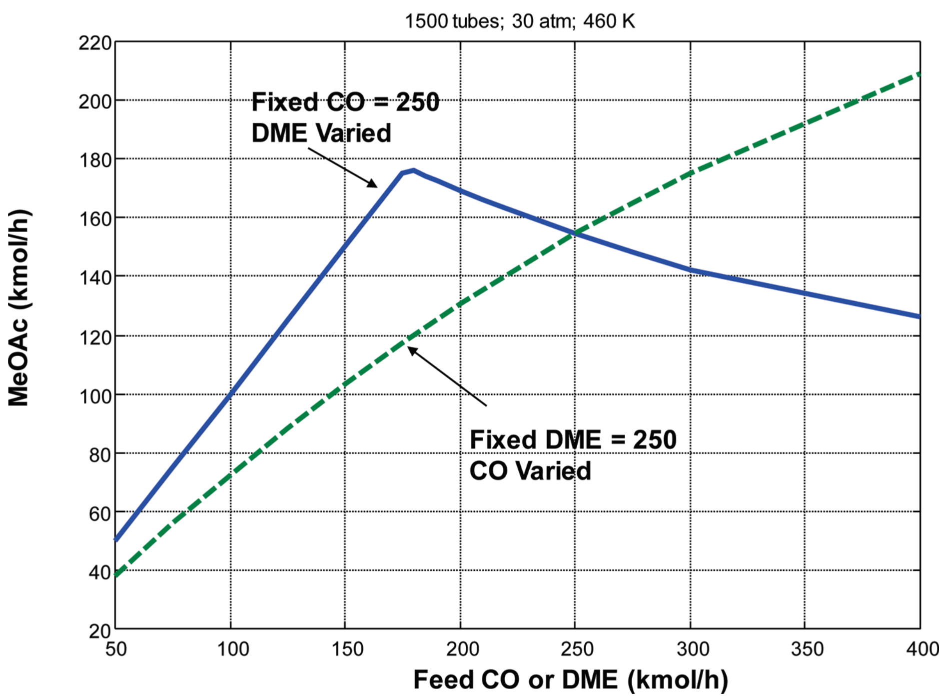

Figure 10. Carbonylation kinetics; effect of fresh feeds.

increasing CO improves methyl acetate production, even when a large excess of CO used.

The solid line gives results for the case when the flow rate of CO is fixed at 250 kmol/h, and the flow rate of DME is varied over a wide range. At low DME flow rates, methyl acetate production increases as more DME is fed. The DME is essentially completely consumed in this region. However, above a certain DME feed rate, methyl acetate production starts to decrease. The kinetics become inhibited at high DME concentrations.

These results indicate that the carbonylation reactor must be run with an excess of CO, not DME. This means that the flowsheet will feature a recycle of gaseous CO, which will involve a recycle compressor.

3.4. Flowsheet Convergence. The carbonylation flowsheet involves two gas recycle streams: vapor from the separator and the vapor distillate from the distillation column. It is also necessary to balance the stoichiometry of the reaction so that the amount of CO fed is exactly equal to the CO consumed in the reaction with the DME feed plus the losses in the purge stream.

Convergence of the recycle loop could not be achieved in Aspen Plus despite numerous approaches to the problem. What did work was a technique[6] suggested many years ago that involves exporting the steady-state Aspen Plus file without the recycle loop closed into Aspen Dynamics as a pressure-driven dynamic simulation. Then the recycle stream is connected, and a control structure is installed to drive the system to the desired steady-state conditions. This control structure may or may not be the same as that used later in plantwide control studies.

The flow rate of DME is flow controlled at 250 kmol/h. The fresh feed of CO into the process is manipulated to maintain a high conversion of DME in the reactor. The flow of the fresh CO is flow controlled by manipulating the work to compressor K_{CO}. The flow controller is “on cascade” with its set point coming from a “conversion controller” whose process variable is 30 kmol/h of DME leaving the reactor. The DME entering the reactor is 278.5 kmol/h, so the specified conversion is 89%.

Other control loops involve holding liquid level in the vaporizer by manipulating heat input, holding separator pressure at 30 atm by manipulating the work to compressor K_1, holding column C3 pressure at 5 atm by manipulating the work to compressor K_2, holding separator temperature by manipulating heat removal in partial condenser HX1, fixing reflux flow in column C3 at 70.8 kmol/h, and holding reflux-drum temperature in column C3 at 300 K by manipulating reboiler heat input. The last two loops maintained the high purity (99.9 mol %) of the methyl acetate bottoms product from the distillation column. Liquid levels in the separator and in the column base are held by manipulating the liquid stream leaving the vessel.

3.5. Optimization. Now that the reactor temperature has been selected from basic kinetic considerations to be at its maximum, the remaining dominant design optimization variables are pressure, reactor size, and purge composition.

3.5.1. Economic Basis. The economics consider both capital investment (compressors, vaporizer, partial condenser, reactor, catalyst, separator tank, distillation column, condenser, and reboiler) and energy costs (compressor work, steam used in the vaporizer, steam used in the column reboiler, and refrigeration used in the column condenser). The two credits are the value of the steam generated in the reactor and the heating value of purge.

Table 2 summarizes the sizing and cost basis used in this analysis. These parameters are taken from Douglas[3] and Turton et al.[4] The compressors use electricity at \$16.8 per GJ. The partial condenser uses cooling water, whose cost is neglected. The vaporizer and reboiler use low-pressure steam at \$7.78 per GJ. The condenser uses 253 K refrigerant at \$7.89 per GJ.

There are two credits in the economics. One is for the steam generated in the reactor. It is low-pressure saturated steam with a value of \$7.78 per GJ. The second is the heating value of the purge stream, which contains mostly hydrogen and CO with small amounts of DME and methyl acetate. The stream can be burned to recover heat. The amount of air to completely combust each component, the component heats of combustion, and the mean heat capacities of the resulting nitrogen, carbon dioxide, and water gas stream are used to find the net heating value of the purge stream (0.259 kJ/kmol). The sensible heat of changing the combustion products from ambient up to a 260 °C stack gas temperature is subtracted from the heat of combustion. A value of \$6 per GJ is used for this fuel, which corresponds to \$1.6 per kmole.

The assessment of economics assumes the DME feed is fixed at 250 kmol/h. The stoichiometric amount of CO would be 250 kmol/h. The incremental amount of CO that must be fed to satisfy reaction and losses is counted as an operating cost. A price of CO fresh feed is assumed to be \$13.2 per kmol (\$370 per tonne). The annual capital cost uses a payback period of 3 years. Total capital investment includes the vaporizer and reactor whose cost is based on heat-transfer area. The size of the separator tank is set by have 10 min of liquid holdup. Other capital investments are the three compressors, the partial condenser, and the distillation column with its vessel, reboiler, and condenser.

The TAC is defined as the annual cost of capital plus the cost of energy (reboiler, vaporizer, refrigerant, and compressor work), plus the cost of CO fresh feed above the 250 kmol/h stoichiometric amount, minus the credits for reactor steam and purge heating value. The amount of methyl acetate produced in all cases is essentially constant at 249.7 kmol/h. This method of economic analysis avoids the necessity of specifying the cost of DME and the value of methyl acetate, which always have quite a bit of uncertainty.

$$\begin{aligned}\text{TAC} = {} & \text{Cost Incremental CO} + \text{Compressor Energy} + \\ & \text{Reboiler Energy} + \text{Vaporizer Energy} + \\ & \text{Condenser Refrigerant} + (\text{Total Capital})/3 - \\ & \text{Credit for Reactor Steam} - \text{Credit for Purge Heating Value}\end{aligned} \tag{8}$$

The optimum number of stages and the feed-stage location in the distillation column are found by minimizing total annual cost. The optimum is a 17-stage column fed on the top tray (Stage 2). This column has a total annual cost of \$403,000 per year compared to a TAC of \$690,000 for a smaller 12-stage column.

3.5.2. Effect of Reactor Size and Purge Composition. The number of tubes in the reactor and the hydrogen composition of the purge stream are dominant design optimization variables. A big reactor requires smaller recycle flow rates, so compression costs are lower. But the capital cost of the reactor vessel and the expensive catalyst are high. Letting the hydrogen concentration build up reduces purge losses and CO fresh feed requirements. But recycle flow rates increase, so compression costs are large.

Figure 11 shows these effects for three different reactor sizes (800, 1000, and 1500 tubes) over a range of purge gas compositions. For all these results, the DME feed flow rate and the methyl acetate product flow rate are essentially fixed. Other fixed variables are separator pressure (30 atm) and temperature (320 K), column pressure (5 atm) and reflux-drum temperature (300 K), DME leaving the reactor (30 kmol/h) and column reflux flow rate (70.8 kmol/h).

The top left graph in Figure 11A shows how recycle flow rates increase with purge hydrogen composition. This occurs because the hydrogen reduces the concentrations of the reactant, so more recycle is required to maintain the high per-pass conversion of DME. As the number of reactor tubes is increased, the required recycle decreases.

The right three graphs in Figure 11A show how the work of the three compressors change with purge concentration and reactor size. The recycle compressor on the separator gas stream (K_1) increases rapidly as purge composition increases but is lowered by using a larger reactor. The compressor on the vapor distillate (K_2) decreases with increasing purge composition because fewer light components are fed to the column at the higher gas flow rates from the separator tank. There is little effect of reactor size.

The fresh CO feed compressor is most strongly affected by reactor size. The separator tank pressure is constant at 30 atm for all these cases, but the higher recycle flow rates result in

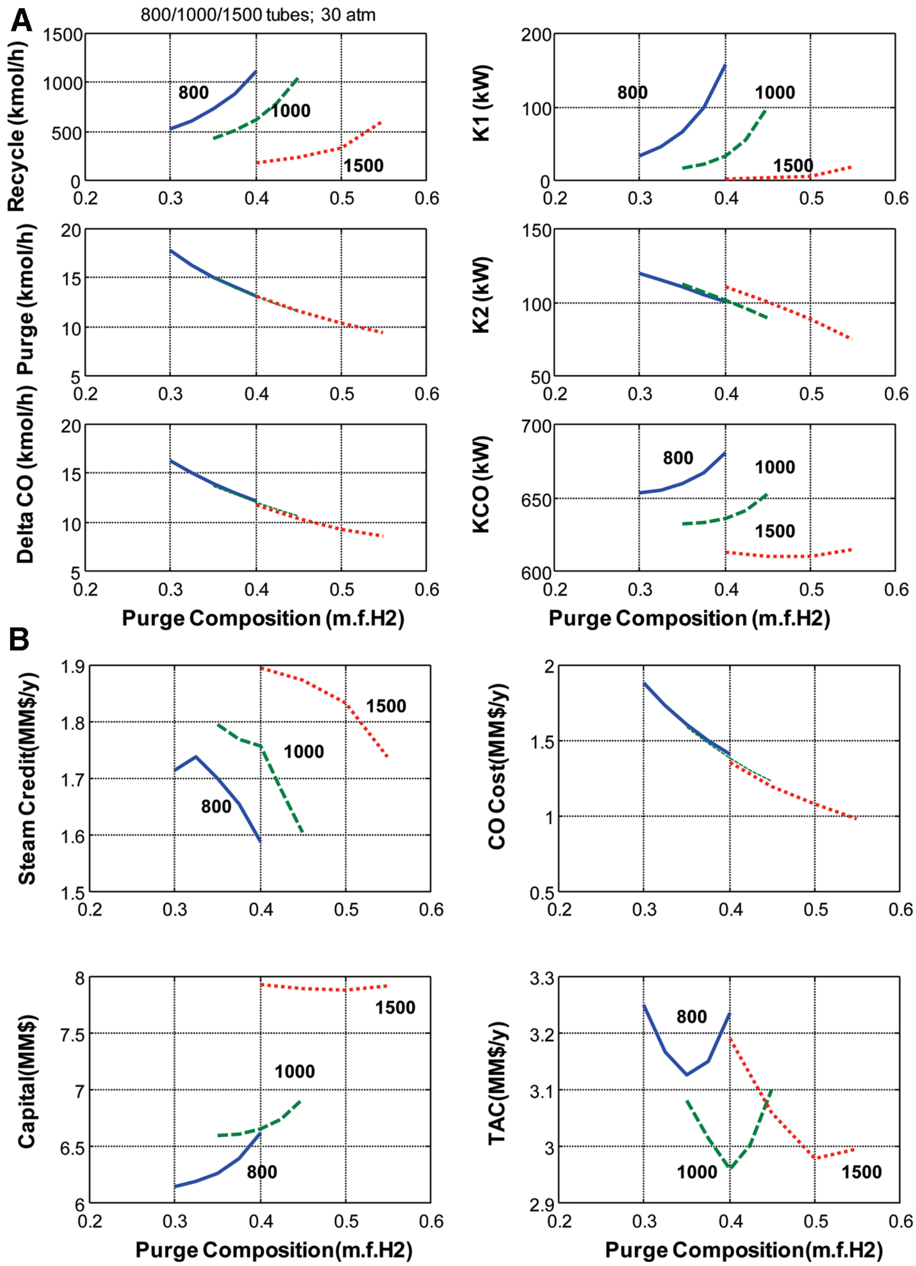

Figure 11. (A) Effect of purge composition and reactor tubes; 30 atm; 460 K; (B) Economic effects of purge composition and reactor tubes.

higher pressure drops through the gas loop (reactor and partial condenser) that raise the discharge pressure of the K_{CO} compressor. The large reactor has lower recycle flow rates and lower pressure drops, so the compressor work is lower and less affected by purge composition.

The middle and bottom graphs in Figure 11A show how less purge and less fresh CO feed is required as purge hydrogen composition is increased, with essentially the same effect for any reactor size. The "Delta CO" variable is the flow rate of CO above the 250 kmol/h stoichiometric amount. Thus increasing purge hydrogen composition lower CO feed costs.

Figure 11B gives economic results. The value of the steam generated in the reactor is shown in the upper left graph. Larger reactors generate more steam because they require lower recycle flow rates. Higher recycle flow rate carry more sensible heat out of the reactor into the partial condenser, whose heat duty increases as recycle flow rate increases. Since higher purge gas hydrogen composition require higher recycle flow rates, reactor steam generation decrease as purge composition increases. A comparison of the top left graph in Figure 11A (showing recycle) with the top left graph in Figure 11B (showing steam) clearly shows this inverse relationship.

The bottom left graph in Figure 11B shows the strong effect of reactor size on total capital cost. This is due to the large contribution of the cost of the reactor vessel ($872,000) and the catalyst ($2,420,000), which are significant components of the total capital cost. In the final design, the capital cost of all three compressors is $2,570,000; the capital cost of the distillation column and associated heat exchangers is $282,000; the capital cost of the separator tank is $200,000; the capital cost of the vaporizer is $120,000; the capital cost of the partial condenser is $300,000. So the reactor cost and the compressor costs dominate the economics of capital investment.

In contrast, the economics of operating costs are dominated by compression costs ($408,000 per year), the cost of excess fresh CO feed ($1,360,000 per year) and the value of the reactor steam ($1,760,000 per year). The energy cost of the reboiler

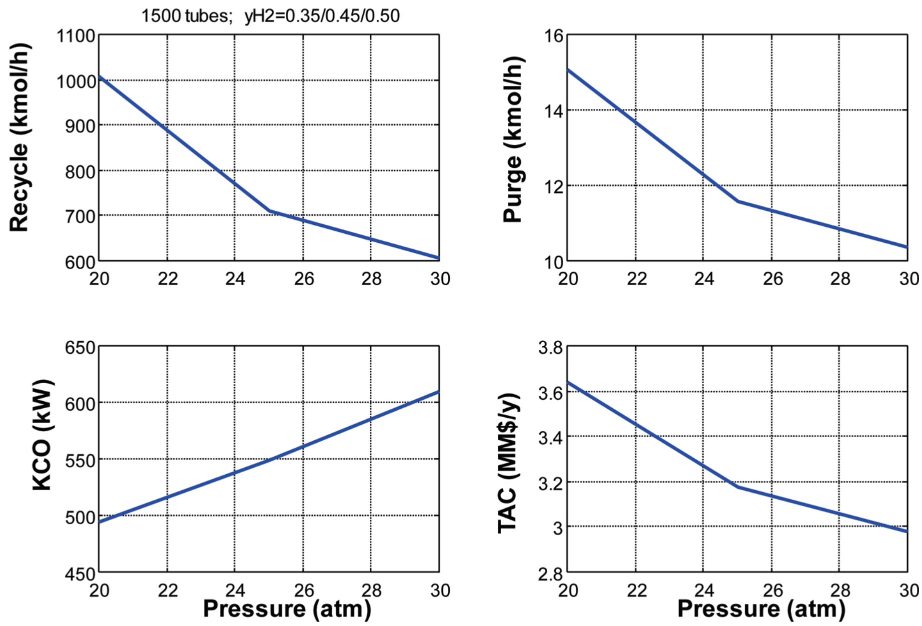

Figure 12. Effect of pressure; 1500 tubes; 460 K.

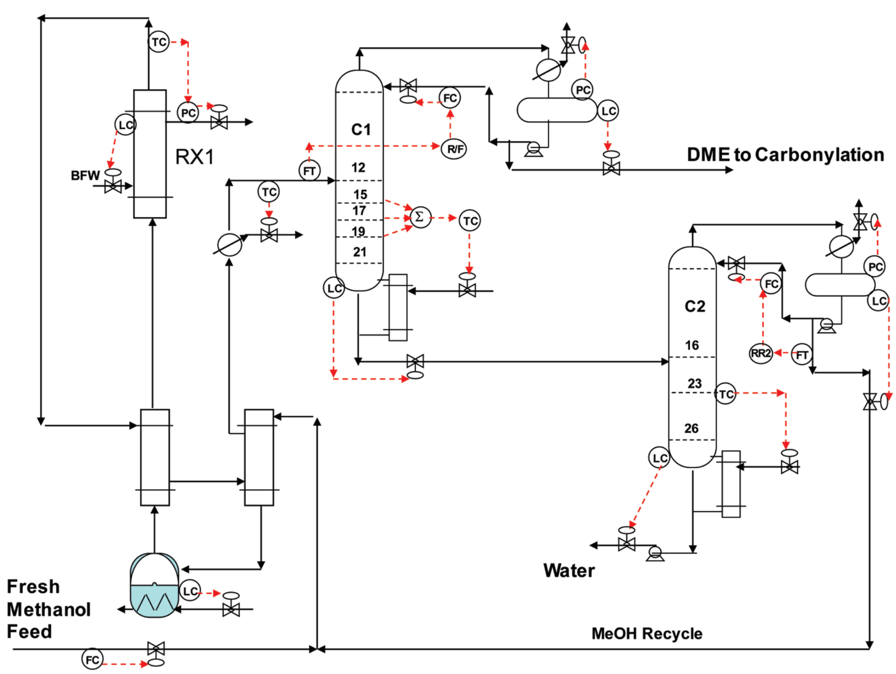

Figure 13. Dehydration control structure.

steam is \$414,000 per year, and the cost of condenser refrigeration is \$163,000 per year.

The effect of reactor size and purge composition on TAC is shown in the bottom right graph of Figure 11B. There is an optimum purge composition for each reactor size that increases as the reactor becomes larger. The minimum TAC design has a reactor with 1000 tubes and a purge composition of 40 mol % hydrogen.

The final important design optimization variable is system pressure. Figure 12 shows that higher pressures reduce the required recycle flow rate and the purge flow rate, which reduce recycle compression costs and the cost of fresh CO feed. However, the compression cost of K_{CO} increases. Since reactant cost is an important variable, TAC continues to decrease as pressure increases. The assumed 30 atm pressure limitation then fixes system pressure. The results shown in Figure 12 are for a 1500 tube reactor with the optimum purge composition for each pressure (35 mol % at 20 atm, 45 mol % at 25 atm, and 50 mol % at 30 atm). A 1500-tube reactor is used because very high recycle flow rates and low purge compositions are needed at the low system pressures).

4. Plantwide Control

Before exporting the steady-state Aspen Plus simulation into Aspen Dynamics, the reflux drums and column bases in the distillation columns are sized to provide 5 min of liquid holdup when at 50% level. The sizes of the reactors and column vessels are known from the steady-state design. The size of the separator is also determined during steady-state design so that its capital investment costs could be calculated. The compressors and heat exchangers are assumed have negligible dynamic lags. The

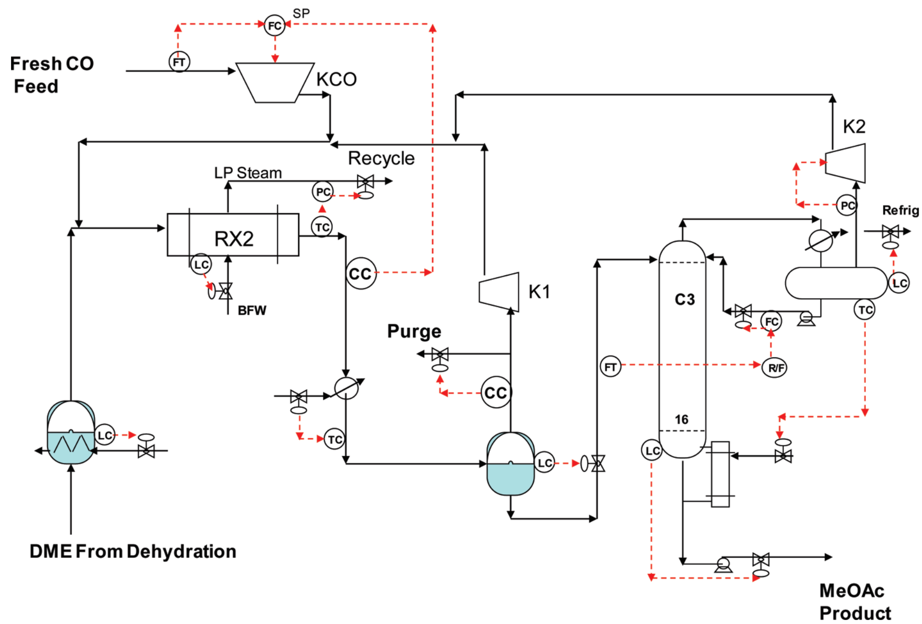

Figure 14. Carbonylation control structure.

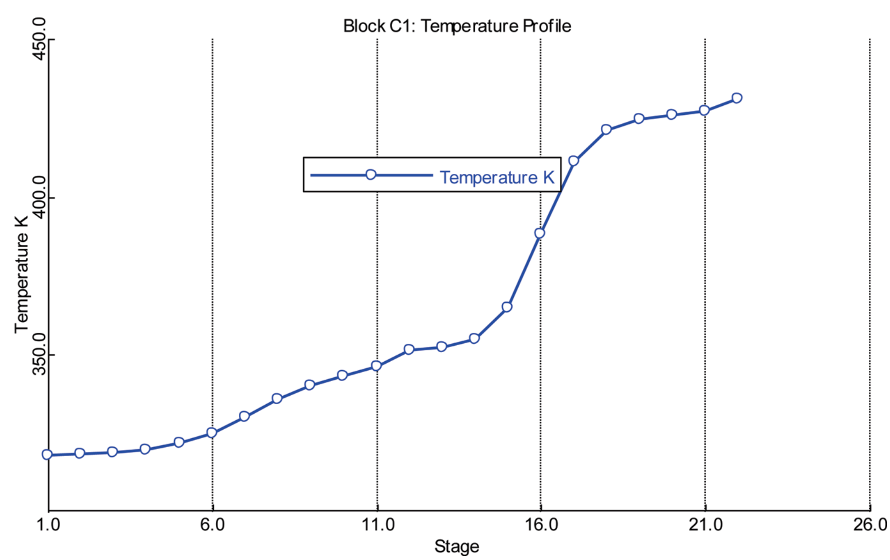

Figure 15. Temperature profile in column C1.

"Gear" numerical integration algorithm is used because it gave much faster computing speed than the default "Implicit Euler" algorithm.

The development of the plantwide control structure presented below is based on the heuristic procedure proposed a decade ago[7] that has been successfully applied to many industrial processes.

4.1. Control Structure. Figures 13 and 14 show the plantwide control structure for the two sections. Conventional PI controllers are used in all loops. All level loops are proportional with $K_C = 2$. The flow controller that manipulates the work of the CO feed compressor uses a gain of 0.5 and an integral time of 0.5 min. The column tray temperature controllers have 1-min deadtimes. The reactor temperature loops have 2-min deadtimes to account for the steam generation dynamics. The composition controllers have 3-min deadtimes. These temperature and composition controllers are tuned by using relay-feedback tests to obtain ultimate gains and periods and then applying Tyreus-Luyben tuning rules.

All of the loops are listed below with their controlled and manipulated variables.

A. Dehydration Section (Figure 13):

1. The fresh feed of liquid methanol is flow controlled.
2. The liquid level in the methanol vaporizer is controlled by manipulating the heat input.
3. The exit temperature of the dehydration reactor RX1 is controlled by manipulating the cooling-medium temperature (pressure of steam).
4. The temperature of the feed to Column C1 is controlled by manipulating cooling water to the heat exchanger.
5. Reflux-to-feed is controlled in C1 by manipulating reflux flow rate.
6. Pressures in C1 and C2 are controlled by manipulating condenser heat removals.
7. Base levels are controlled in C1 and C2 by manipulating bottoms flow rates.
8. Reflux-drum levels in C1 and C2 are controlled by manipulating distillate flow rates.

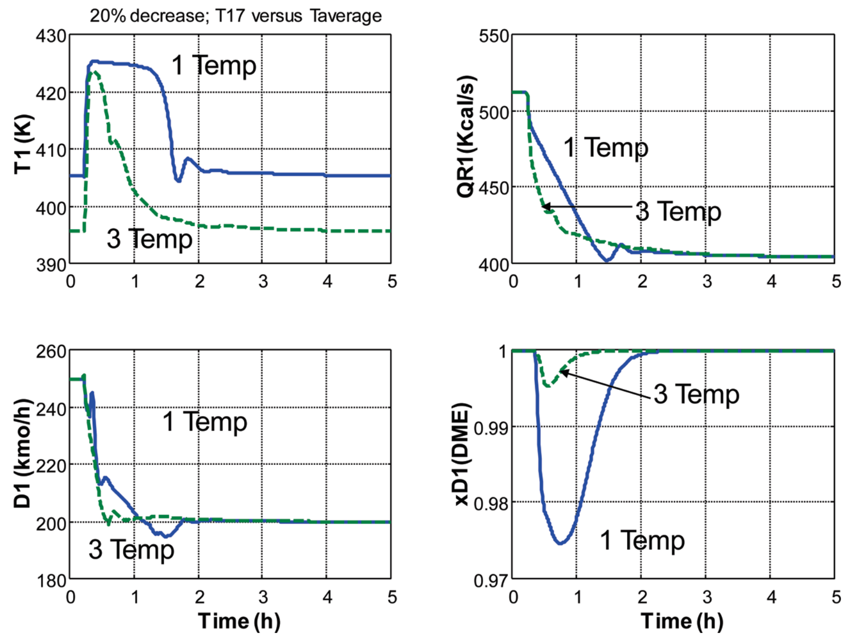

Figure 16. Average temperature control of column C1.

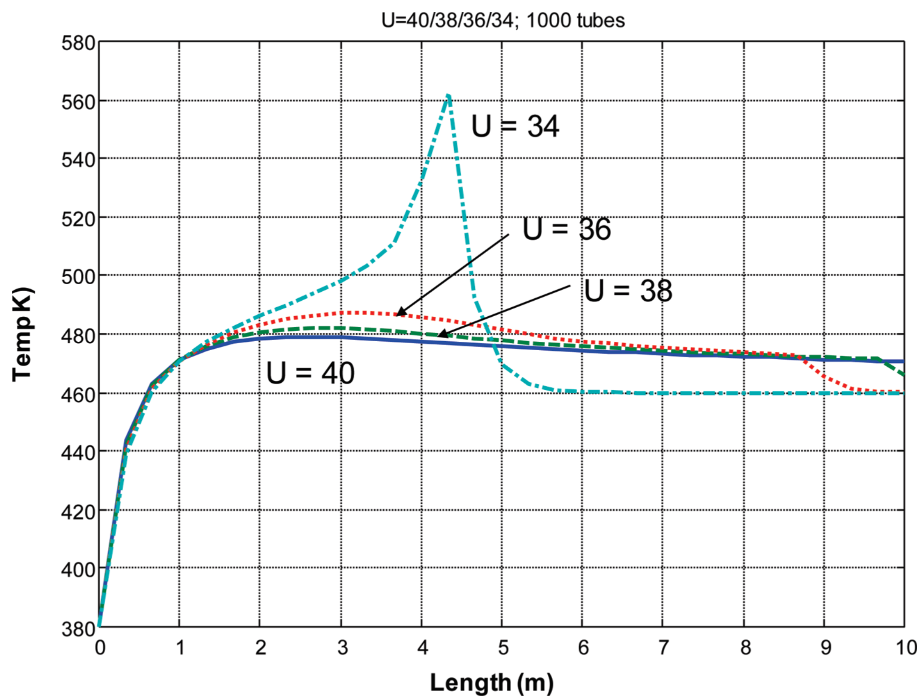

Figure 17. Carbonylation temperature profiles.

9. An average temperature in C1 (Stages 15, 17 and 19) is controlled by manipulating reboiler heat input.
10. The reflux ratio in C2 is controlled by manipulating reflux flow rate.
11. The temperature on Stage 23 in C2 is controlled by manipulating reboiler heat input.

B. Carbonylation Section (Figure 14):

1. The liquid level in the DME vaporizer is controlled by manipulating the heat input.
2. The exit temperature of the carbonylation reactor RX2 is controlled by manipulating the cooling-medium temperature (pressure of steam).
3. The composition of DME in the effluent of the reactor is controlled by manipulating the fresh feed of CO through a cascade flow controller whose output signal manipulates the work to compressor KCO.
4. The temperature of the feed to the separator tank is controlled by manipulating cooling water to the heat exchanger.
5. The liquid level in the separator is controlled by manipulating the feed to Column C3.
6. The hydrogen composition of the gas from the separator is controlled at 40 mol % hydrogen by manipulating the gas purge.
7. The work to compressor K1 is fixed.
8. The temperature in the reflux drum of Column C3 is controlled by manipulating reboiler heat input.
9. The reflux in Column C3 is ratioed to the feed.

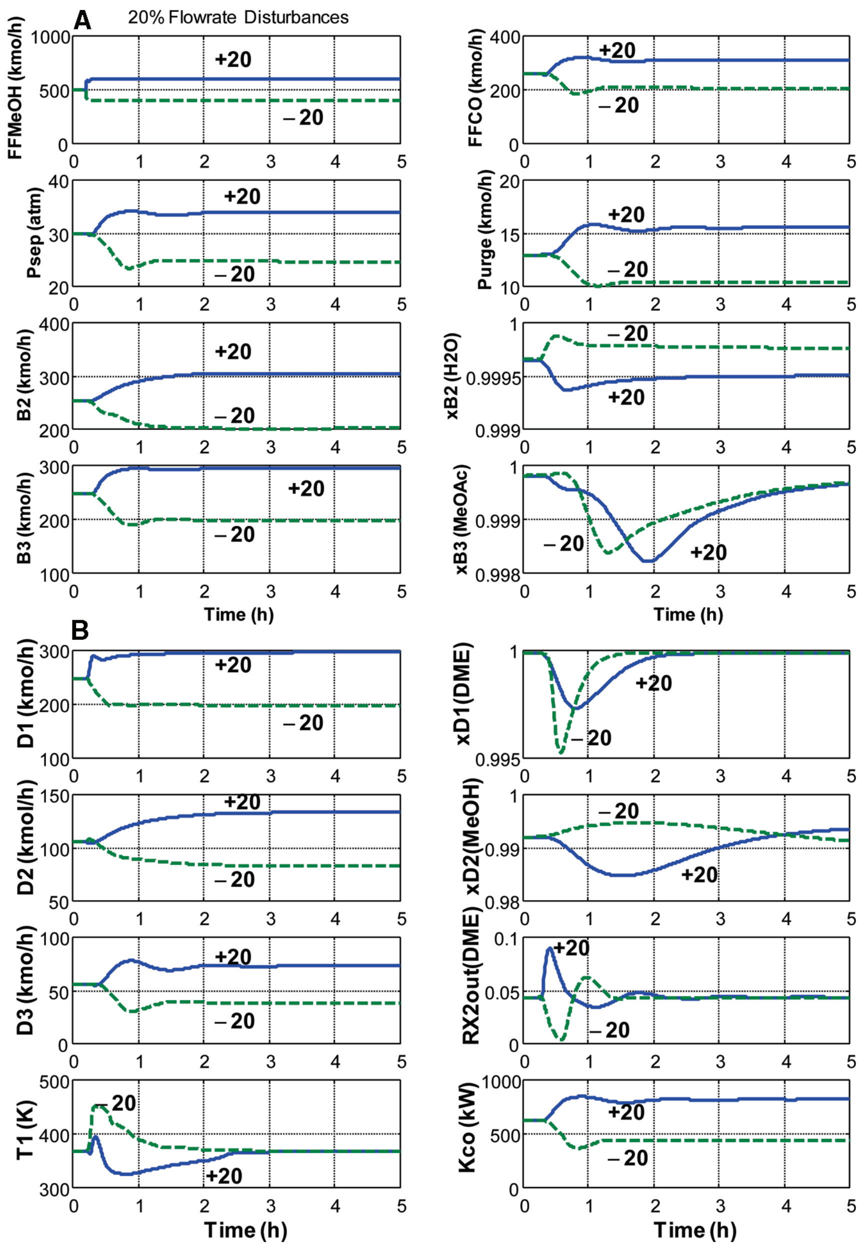

Figure 18. (A) 20% flow rate disturbances, (B) continued.

10. The base level in C3 is controlled by manipulating bottoms flow rate.
11. The pressure in C3 is controlled by manipulating work to Compressor K2.
12. The reflux-drum level in C3 is controlled by manipulating condenser heat removal.

There are several key loops in this structure that are discussed in detail below. Notice that there is no control of the pressure in the gas loop in the carbonylation section. Pressure floats up and down with the load on the plant to force the complete consumption of both reactants.

4.1.1. Average Temperature Control in Column C1. The temperature difference between the top and the bottom of Column C1 is quite large (320 to 430 K) because the DME distillate is much lighter than the other components (Figure 15). The steep part of the temperature profile is in the stripping section. In this situation the use of an average temperature control structure instead of a single temperature structure is often used to improve controllability.

Figure 16 demonstrates this improvement. The solid lines are when the temperature on Stage 17 is controlled. The dashed lines are when the sum of the temperatures on Stages 15, 17, and 19 are controlled. The disturbance is a 20% decrease in the fresh feed of methanol to the plant. Note that since Aspen Dynamics uses metric units, the "average" temperature (actually the sum of the three temperatures) is Celsius. The single Stage 17 temperature is in Kelvin.

The big improvement is in the purity of the DME distillate. With single temperature control, a large transient drop in purity results (bottom right graph in Figure 16) down from 99.9 mol % down to 97.5 mol % DME. With average temperature control, the transient drop in purity is greatly reduced.

4.1.2. R/F Control in Column C1 and RR in Column C2. Temperature control is used in both columns, but to decide how the other control degree of freedom should be selected, a "feed composition sensitivity analysis" is performed on both columns. The two design specifications in each column are fixed and feed compositions of the light and heavy key components are varied around their design values. The required changes in reflux flow rate and reflux ratio are determined.

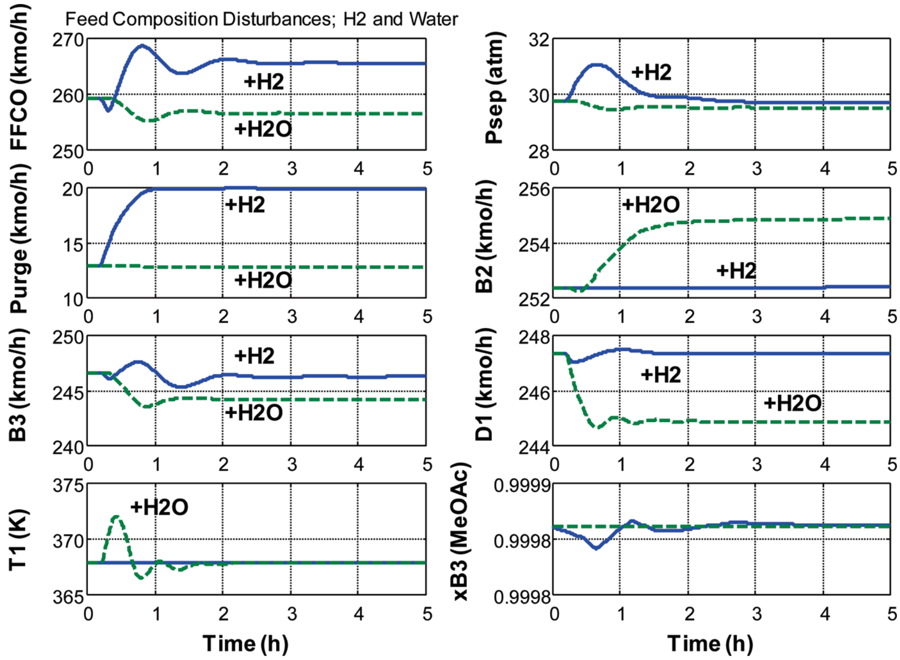

Figure 19. Feed composition disturbances.

In Column C1 the required changes in the reflux-to-feed ratio (R/F) are much smaller than the required changes in reflux ratio (RR). Therefore the R/F structure is selected.

In Column C2 the required changes in reflux ratio are smaller than the required changes in the reflux-to-feed ratio. Therefore the RR structure is selected.

4.1.3. Conversion in Carbonylation Reactor. The most important loop in the whole plant is the one that controls the concentration of DME in the reactor effluent leaving the carbonylation reactor. If DME is increasing, more fresh feed of CO is added.

This structure requires an online measurement of composition. Several other alternative structures were explored that avoided composition measurement, but none was found that could handle the difficult problem of precisely balancing the addition rate of CO to satisfy the stoichiometry of the reaction.

Many plants with a gas fresh feed can be successfully controlled by bringing in the gas on pressure control. If more reactant is being fed than consumed, the pressure will rise, and the pressure control cuts back on the fresh gas feed. This technique eliminates a composition measurement.

However, in this process there are two gas streams into the carbonylation reactor. If pressure is increasing, it can be due to too much or too little CO. This control structure did not provide stable regulatory control.

4.1.4. Control Structure in Column C3. Because of the high pressure in the separator, the liquid phase contains quite a bit of lighter components in addition to the product methyl acetate. The liquid composition is 10.76 mol % CO and 5.79 mol % DME, and these light components must be removed and recycled back to the reactor. The C3 distillation column has this job.

The very light CO and DME would require a very high pressure if cooling water were used in the condenser, so refrigeration is used. To keep the reflux-drum temperature from being very low, some methyl acetate is taken overhead. The distillate is taken as a vapor from the top of the reflux drum and is compressed and recycled.

A simple control structure was found to be quite effective since the basic separation is an easy one (few trays and modest reflux ratio). The reflux flow rate is ratioed to the feed flow rate. Reflux-drum temperature is controlled at 300 K (so medium cold refrigerant can be used) by manipulating reboiler duty. Pressure is controlled by manipulating Compressor K2 work. Reflux-drum level is controlled by manipulating condenser refrigerant duty.

4.1.5. Control Structure for Reactors. The exit temperatures of the two cooled reactors are controlled by manipulating the cooling medium temperature. Physically this corresponds to manipulating the pressure of the steam being generated. Temperature control of the dehydration reactor was easy to set up and provided effective control.

Temperature control of the carbonylation reactor was found to be quite difficult. Relay-feedback test could not be run because of numerical integration errors. To overcome these problems, the value of the overall heat-transfer coefficient was increased from the design value of 0.28 to 0.40 kW K^{-1} m^{-2}. Figure 17 shows the strong dependence of the temperature profile on the value of the overall heat-transfer coefficient. These numerical problems may indicate real physical problems in this highly exothermic system. This issue is under study.

4.2. Dynamic Performance. The coupled system was subjected to large disturbances throughput and feed compositions. Figure 18 gives the responses of several important variables to 20% disturbances in the set point of the fresh methanol flow controller. The solid lines are 20% increases, and the dashed lines are 20% decreases. Stable regulatory control is achieved for these large disturbances.

The second graph from the top on the left of Figure 18A shows that the pressure in the separator (Psep) floats up and down with throughput. Higher feed rates require higher pressures, which results in higher power consumption in the fresh CO compressor. The production rates of methyl acetate (B3) and water (B2) change directly with throughput, as does the flow rate of the purge stream.

The purities of these streams [$x_{B3(MeOAc)}$ and $x_{B2(H2O)}$] are held quite close to their specifications. The top right graph in Figure 18B shows that the purity of the DME ($x_{D1(DME)}$) drops slightly for both disturbances, which puts more methanol into the carbonylation section and causes the small transient drop in the methyl acetate purity.

Figure 19 shows how disturbances in the compositions of the two fresh feeds affect the system. The solid lines are when the hydrogen impurity in the CO increases from 2 to 3 mol %. The major effect, as expected, is an increase in the purge flow rate. The flow rate of fresh CO also must increase since the amount of fresh methanol fed and the resulting DME produced require the same amount of CO to react and the CO losses are larger.

The dashed lines are when the water impurity in the fresh methanol increases from 1 to 2 mol %. Since slightly less methanol is being fed, there is a small reduction in the DME produced (shown as D1 in the right graph, third from the top). Naturally slightly less methyl acetate is produced (B3) and slightly more water is produced (B2).

5. Conclusion

The design and control of a fairly complex chemical process has been studied. The two sections of the plants have their individual reaction and separations sections with a recycle stream in each. Steady-state design involves a number of economic trade-offs between the many design optimization variables. Reactor size, temperature, and pressure affect separation and compression costs, as well as reactant losses in a purge stream.

A plantwide control structure is developed and demonstrated to provide effective control for large disturbances.

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